LIFE EXPECTANCY DATA ANALYSIS

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Introduction and Our Dataset

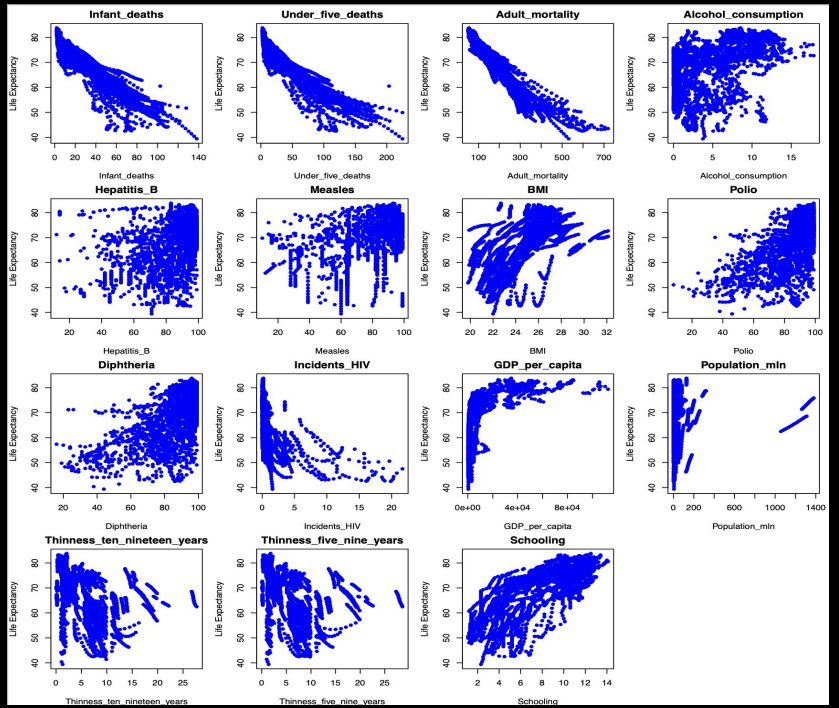
- **Unit of Analysis:** 179 countries from 2000-2015, (2,864 rows)
- **21 Variables on:** life expectancy, health, immunization, and economic and demographic information
- **Primary question:** What are the most decisive predictive factors influencing global life expectancy?

```
> who_data <- read.csv("Desktop/Life-Expectancy-Data-Updated.csv")
> head(who_data)
     Country                       Region Year Infant_deaths Under_five_deaths Adult_mortality Alcohol_consumption Hepatitis_B Measles  BMI Polio Diphtheria Incidents_HIV GDP_per_capita
1    Turkiye                  Middle East 2015          11.1              13.0        105.8240                1.32          97      65 27.8    97          97         0.08          11006
2      Spain                European Union 2015           2.7               3.3         57.9025               10.35          97      94 26.0    97          97         0.09          25742
3      India                         Asia 2007          51.5              67.9        201.0765                1.57          60      35 21.2    67          64         0.13           1076
4     Guyana                 South America 2006          32.8              40.5        222.1965                5.68          93      74 25.3    92          93         0.79           4146
5     Israel                  Middle East 2012           3.4               4.3         57.9510                2.89          97      89 27.0    94          94         0.08          33995
6 Costa Rica Central America and Caribbean 2006           9.8              11.2         95.2200                4.19          88      86 26.4    89          89         0.16           9110
  Population_mln Thinness_ten_nineteen_years Thinness_five_nine_years Schooling Economy_status_Developed Economy_status_Developing Life_expectancy
1          78.53                         4.9                       4.8       7.8                       0                         1             76.5
2          46.44                         0.6                       0.5       9.7                       1                         0             82.8
3        1183.21                        27.1                      28.0       5.0                       0                         1             65.4
4           0.75                         5.7                       5.5       7.9                       0                         1             67.0
5           7.91                         1.2                       1.1      12.8                       1                         0             81.7
6           4.35                         2.0                       1.9       7.9                       0                         1             78.2
```

Preliminary Data Exploration

- Removing Unnecessary Variables
  - Country, Region, Year,
  - Economy_status_Developing/Economy_status_Developed

```
> cor(data$Infant_deaths, data$Life_expectancy)
[1] -0.9200319
> cor(data$Under_five_deaths, data$Life_expectancy)
[1] -0.9204191
> cor(data$Adult_mortality, data$Life_expectancy)
[1] -0.9453604
>
> cor(data$GDP_per_capita, data$Life_expectancy)
[1] 0.5830897
> cor(data$Diphtheria, data$Life_expectancy)
[1] 0.6275414
>
> cor(data$Population_mln, data$Life_expectancy)
[1] 0.02629788
```



VIF

```
> vif(reduced_first_model)
                 Infant_deaths               Under_five_deaths                Adult_mortality
                      44.329144                      44.469991                       7.618888
         Alcohol_consumption                    Hepatitis_B                        Measles
                       1.939630                       2.560952                       1.587422
                           BMI                         Polio                    Diphtheria
                       2.627136                      11.990081                      12.940367
                 Incidents_HIV                GDP_per_capita                Population_mln
                       2.890280                       2.016715                       1.150097
Thinness_ten_nineteen_years    Thinness_five_nine_years                      Schooling
                       8.949993                       8.936053                       4.273524
```

The VIF test suggests there is multicollinearity with **infant deaths, under 5 deaths, diphtheria, and polio (>10)**, and after cross-checking with our correlation matrix we removed these variables.

```
> vif(reduced_first_model)
     Alcohol_consumption                 Hepatitis_B                     Measles                         BMI                 Incidents_HIV
                1.823265                    1.299219                    1.530168                    2.202095                    1.093900
           GDP_per_capita                Population_mln Thinness_ten_nineteen_years Thinness_five_nine_years                Schooling
                1.568157                    1.123121                    8.920279                    8.833766                    3.320544
>
```

Much better (VIF<10)!

Cross check with
Correlation matrix

```
> cor(data)
                  Infant_deaths Under_five_deaths Adult_mortality Alcohol_consumption Hepatitis_B     Measles
Infant_deaths        1.00000000       0.985651346      0.79466086          -0.45452615 -0.51256224 -0.52628201
Under_five_deaths    0.98565135       1.000000000      0.80236112          -0.40936740 -0.50742741 -0.51297174
Adult_mortality      0.79466086       0.802361123      1.00000000          -0.24479376 -0.34488221 -0.41615254
Alcohol_consumption -0.45452615      -0.409367397     -0.24479376           1.00000000  0.16843582  0.31860293
Hepatitis_B         -0.51256224      -0.507427407     -0.34488221           0.16843582  1.00000000  0.42916779
Measles             -0.52628201      -0.512971742     -0.41615254           0.31860293  0.42916779  1.00000000
BMI                 -0.66198827      -0.665255042     -0.52286551           0.28403195  0.34542091  0.41632141
Polio               -0.74079046      -0.742983474     -0.52422554           0.30192623  0.72434526  0.51409629
Diphtheria          -0.72187465      -0.725355032     -0.51380270           0.29901592  0.76178009  0.49405877
Incidents_HIV        0.34945826       0.369617726      0.69911938          -0.03411801 -0.07578195 -0.15058000
GDP_per_capita      -0.51228611      -0.469681668     -0.51012141           0.44396595  0.15937504  0.31372372
Population_mln       0.00762199      -0.005234231     -0.05384768          -0.03911866 -0.08239640 -0.09822189
Thinness_ten_nineteen_years  0.49119174  0.466978458  0.38214030          -0.44636618 -0.20845350 -0.34070533
Thinness_five_nine_years     0.47763934  0.450755699  0.37979229          -0.43302972 -0.21379442 -0.36696995
Schooling           -0.78851253      -0.773195983     -0.58103548           0.61572804  0.34764345  0.49839128
Life_expectancy     -0.92003192      -0.920419134     -0.94536036           0.39915911  0.41780443  0.49001859
                          BMI       Polio  Diphtheria Incidents_HIV GDP_per_capita Population_mln
Infant_deaths       -0.6619883 -0.74079046 -0.72187465   0.34945826    -0.51228611    0.007621990
Under_five_deaths   -0.6652550 -0.74298347 -0.72535503   0.36961773    -0.46968167    -0.005234231
Adult_mortality     -0.5228655 -0.52422554 -0.51380270   0.69911938    -0.51012141    -0.053847680
Alcohol_consumption  0.2840319  0.30192623  0.29901592  -0.03411801     0.44396595    -0.039118659
Hepatitis_B          0.3454209  0.72434526  0.76178009  -0.07578195     0.15937504    -0.082396398
Measles              0.4163214  0.51409629  0.49405877  -0.15058000     0.31372372    -0.098221891
BMI                  1.0000000  0.45720604  0.42650090  -0.16114208     0.33617960    -0.166482004
Polio                0.4572060  1.00000000  0.95317790  -0.14795220     0.31378567    -0.033485888
Diphtheria           0.4265009  0.95317790  1.00000000  -0.14693191     0.31332094    -0.027335977
Incidents_HIV       -0.1611421 -0.14795220 -0.14693191   1.00000000    -0.16958972    -0.058039708
GDP_per_capita       0.3361796  0.31378567  0.31332094  -0.16958972     1.00000000    -0.040838867
Population_mln      -0.1664820 -0.03348589 -0.02733598  -0.05803971    -0.04083887     1.000000000
Thinness_ten_nineteen_years -0.5964833 -0.31268545 -0.30446625  0.18876454  -0.37526974  0.256322009
Thinness_five_nine_years -0.5991122 -0.30699811 -0.29559745  0.19384734  -0.38103211  0.258485836
Schooling            0.6354752  0.55276511  0.53562097  -0.20124620     0.58062592    -0.033561816
Life_expectancy      0.5984233  0.64121746  0.62754139  -0.55302746     0.58308972    0.026297880
                  Thinness_ten_nineteen_years Thinness_five_nine_years   Schooling Life_expectancy
Infant_deaths                       0.4911917                0.4776393 -0.78851253     -0.92003192
Under_five_deaths                   0.4669785                0.4507557 -0.77319598     -0.92041913
Adult_mortality                     0.3821403                0.3797923 -0.58103548     -0.94536036
Alcohol_consumption                -0.4463662               -0.4330297  0.61572804      0.39915911
Hepatitis_B                        -0.2084535               -0.2137944  0.34764345      0.41780443
Measles                            -0.3407053               -0.3669700  0.49839128      0.49001859
BMI                                -0.5964833               -0.5991122  0.63547517      0.59842332
Polio                              -0.3126855               -0.3069981  0.55276511      0.64121746
Diphtheria                         -0.3044662               -0.2955975  0.53562097      0.62754139
Incidents_HIV                       0.1887645                0.1938473 -0.20124620     -0.55302746
GDP_per_capita                     -0.3752697               -0.3810321  0.58062592      0.58308972
Population_mln                      0.2563220                0.2584858 -0.03356182      0.02629788
Thinness_ten_nineteen_years         1.0000000                0.9387571 -0.57148516     -0.46782450
Thinness_five_nine_years            0.9387571                1.0000000 -0.55137635     -0.45816623
Schooling                          -0.5714852               -0.5513764  1.00000000      0.73248447
Life_expectancy                    -0.4678245               -0.4581662  0.73248447      1.00000000
```

Variables after VIF

- **Dependent variable**: Life_expectancy
- **Independent variables (10)**: Alcohol_consumption, Hepatitis_B, Measles, BMI, Incidents_HIV, GDP_per_capita, Population_mln, Thinness_ten_nineteen_years, Thinness_five_nine_years, Schooling

01

First Order Model

1. **First Order Model**

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9 + \beta_{10} x_{10}$$

Note that: E(y) = Life expectancy, x_1 = Alcohol Consumption, x_2 = % of Hepatitis B immunization coverage, x_3 = % of Measles immunization coverage,

x_4 = BMI, x_5 = Incidents of HIV per 1000 population, x_6 = GDP per capita, x_7 = Population in millions, x_8 = Thinness (10-19 years), x_9 = Thinness (5-9 years), x_{10}= Schooling Years

```
Call:
lm(formula = Life_expectancy ~ Alcohol_consumption + Hepatitis_B +
    Measles + BMI + Incidents_HIV + GDP_per_capita + Population_mln +
    Thinness_ten_nineteen_years + Thinness_five_nine_years +
    Schooling, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-16.7534  -2.9282   0.1882   3.2362  14.7943

Coefficients:
                               Estimate Std. Error t value Pr(>|t|)
(Intercept)                   2.822e+01  1.305e+00  21.617  < 2e-16 ***
Alcohol_consumption          -7.504e-03  2.696e-02  -0.278   0.7807
Hepatitis_B                   8.928e-02  5.665e-03  15.760  < 2e-16 ***
Measles                       3.454e-02  5.270e-03   6.555 6.58e-11 ***
BMI                           8.901e-01  5.377e-02  16.555  < 2e-16 ***
Incidents_HIV                -1.574e+00  3.491e-02 -45.071  < 2e-16 ***
GDP_per_capita                1.290e-04  5.878e-06  21.945  < 2e-16 ***
Population_mln                4.438e-03  6.173e-04   7.191 8.21e-13 ***
Thinness_ten_nineteen_years  -1.110e-01  5.350e-02  -2.075   0.0381 *
Thinness_five_nine_years      2.123e-01  5.221e-02   4.067 4.90e-05 ***
Schooling                     9.759e-01  4.567e-02  21.367  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.254 on 2853 degrees of freedom
Multiple R-squared:  0.7962,    Adjusted R-squared:  0.7955
F-statistic:  1115 on 10 and 2853 DF,  p-value: < 2.2e-16
```

$$E(y) = 85.07 - 0.05152x_1 - 0.05295x_2 - 0.004893x_3 + 0.09023x_4 - 0.009262x_5 + 0.001228x_6 - 0.1674x_7 + 0.0008691x_8 + 0.003257x_9 + 0.1005x_{10} + 0.00003043x_{11} - 0.0001814x_{12} - 0.03313x_{13} - 0.002449x_{14} + 0.1119x_{15}$$

$$H_0: \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = \beta_7 = \beta_8 = \beta_9 = \beta_{10} = \beta_{11} = \beta_{12} = \beta_{13} = \beta_{14} = \beta_{15}$$

$$H_0: \text{at least one } B_i \text{ where } i = 1, 2, 3, \ldots, 15$$

$\alpha = 0.05$

$P = 2.2e\text{-}16 < 0.05$

Reject H0, can conclude the model is useful and proceed!

02

Stepwise Regression

Forward Stepwise Regression

```
Step:  AIC=8301.76
Life_expectancy ~ Schooling + Incidents_HIV + Hepatitis_B + GDP_per_capita +
    BMI + Population_mln + Measles + Thinness_five_nine_years +
    Thinness_ten_nineteen_years

                      Df Sum of Sq   RSS    AIC
<none>                             51619 8301.8
+ Alcohol_consumption  1     1.402 51618 8303.7
```

Variable that was removed: **Alcohol consumption**

Stepwise Conclusion

- Our stepwise forward shows 9/10 of our starting variables to be statistically significant:
  - Both stepwise regressions removed **alcohol consumption** as a variable
  - After checking the p-value (0.7807) from our model summary, we decided **not** to include it in our parsimonious model

- Our new first order model:

```
full_model <- lm(Life_expectancy ~  Hepatitis_B + Measles + BMI  +
                 Incidents_HIV + GDP_per_capita + Population_mln +
                   Thinness_ten_nineteen_years + Thinness_five_nine_years + Schooling, data = data)
```

```
Call:
lm(formula = Life_expectancy ~ Alcohol_consumption + Hepatitis_B +
    Measles + BMI + Incidents_HIV + GDP_per_capita + Population_mln +
    Thinness_ten_nineteen_years + Thinness_five_nine_years +
    Schooling, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-16.7534  -2.9282   0.1882   3.2362  14.7943

Coefficients:
                             Estimate Std. Error t value Pr(>|t|)
(Intercept)                 2.822e+01  1.305e+00  21.617  < 2e-16 ***
Alcohol_consumption        -7.504e-03  2.696e-02  -0.278   0.7807
Hepatitis_B                 8.928e-02  5.665e-03  15.760  < 2e-16 ***
Measles                     3.454e-02  5.270e-03   6.555 6.58e-11 ***
BMI                         8.901e-01  5.377e-02  16.555  < 2e-16 ***
Incidents_HIV              -1.574e+00  3.491e-02 -45.071  < 2e-16 ***
GDP_per_capita              1.290e-04  5.878e-06  21.945  < 2e-16 ***
Population_mln              4.438e-03  6.173e-04   7.191 8.21e-13 ***
Thinness_ten_nineteen_years -1.110e-01  5.350e-02  -2.075   0.0381 *
Thinness_five_nine_years    2.123e-01  5.221e-02   4.067 4.90e-05 ***
Schooling                   9.759e-01  4.567e-02  21.367  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 4.254 on 2853 degrees of freedom
Multiple R-squared:  0.7962,    Adjusted R-squared:  0.7955
F-statistic:  1115 on 10 and 2853 DF,  p-value: < 2.2e-16
```

Final Variables

- **Dependent variable**: Life_expectancy
- **Independent variables (9)**:
  Hepatitis_B, Measles, BMI, Incidents_HIV, GDP_per_capita, Population_mln, Thinness_ten_nineteen_years, Thinness_five_nine_years, Schooling

03

Comparing Models (Four in total)

Model 1: First order complete

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9$$

Note that: E(y) = Life expectancy, x_1 = % of Hepatitis B immunization coverage, x_2 = % of Measles immunization coverage,

x_3 = BMI, x_4 = Incidents of HIV per 1000 population, x_5 = GDP per capita, x_6 = Population in millions,

x_7 = Thinness (10-19 years), x_8 = Thinness (5-9 years), x_9= Schooling Years

Model 2: Second Order with Only Interaction Terms

2nd order with only interactions

$$
\begin{aligned}
E(y) = {} & \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9 + \\
& \beta_{10} x_1 x_2 + \beta_{11} x_1 x_3 + \beta_{12} x_1 x_4 + \beta_{13} x_1 x_5 + \beta_{14} x_1 x_6 + \beta_{15} x_1 x_7 + \beta_{16} x_1 x_8 + \beta_{17} x_1 x_9 + \\
& \beta_{18} x_2 x_3 + \beta_{19} x_2 x_4 + \beta_{20} x_2 x_5 + \beta_{21} x_2 x_6 + \beta_{22} x_2 x_7 + \beta_{23} x_2 x_8 + \beta_{24} x_2 x_9 + \\
& \beta_{25} x_3 x_4 + \beta_{26} x_3 x_5 + \beta_{27} x_3 x_6 + \beta_{28} x_3 x_7 + \beta_{29} x_3 x_8 + \beta_{30} x_3 x_9 + \\
& \beta_{31} x_4 x_5 + \beta_{32} x_4 x_6 + \beta_{33} x_4 x_7 + \beta_{34} x_4 x_8 + \beta_{35} x_4 x_9 + \\
& \beta_{36} x_5 x_6 + \beta_{37} x_5 x_7 + \beta_{38} x_5 x_8 + \beta_{39} x_5 x_9 + \\
& \beta_{40} x_6 x_7 + \beta_{41} x_6 x_8 + \beta_{42} x_6 x_9 + \\
& \beta_{43} x_7 x_8 + \beta_{44} x_7 x_9 + \\
& \beta_{45} x_8 x_9
\end{aligned}
$$

```
#second order with only interactions

model3 <- lm(Life_expectancy ~ Hepatitis_B + Measles + BMI + Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_nineteen_years + Thinness_five_nine_years + Schooling +
               Hepatitis_B*Measles + Hepatitis_B*BMI + Hepatitis_B*Incidents_HIV + Hepatitis_B*GDP_per_capita + Hepatitis_B*Population_mln + Hepatitis_B*Thinness_ten_nineteen_years + Hepatitis_B*Thinness_five_nine_years + Hepatitis_B*Schooling +
               Measles*BMI + Measles*Incidents_HIV + Measles*GDP_per_capita + Measles*Population_mln + Measles*Thinness_ten_nineteen_years + Measles*Thinness_five_nine_years + Measles*Schooling +
               BMI*Incidents_HIV + BMI*GDP_per_capita + BMI*Population_mln + BMI*Thinness_ten_nineteen_years + BMI*Thinness_five_nine_years + BMI*Schooling +
               Incidents_HIV*GDP_per_capita + Incidents_HIV*Population_mln + Incidents_HIV*Thinness_ten_nineteen_years + Incidents_HIV*Thinness_five_nine_years + Incidents_HIV*Schooling +
               GDP_per_capita*Population_mln + GDP_per_capita*Thinness_ten_nineteen_years + GDP_per_capita*Thinness_five_nine_years + GDP_per_capita*Schooling +
               Population_mln*Thinness_ten_nineteen_years + Population_mln*Thinness_five_nine_years + Population_mln*Schooling +
               Thinness_ten_nineteen_years*Thinness_five_nine_years + Thinness_ten_nineteen_years*Schooling,
             data = data)
```

Model3: Second Order with Squared Terms

2nd order with only with squared terms

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9 + \beta_{10} x_1^2 + \beta_{11} x_2^2 + \beta_{12} x_3^2 + \beta_{13} x_4^2 + \beta_{14} x_5^2 + \beta_{15} x_6^2 + \beta_{16} x_7^2 + \beta_{17} x_8^2 + \beta_{18} x_9^2$$

```
#second order with only squared terms
model2 <- lm(Life_expectancy~Hepatitis_B+Measles+BMI+Incidents_HIV+GDP_per_capita+Population_mln+Thinness_ten_nineteen_years+Thinness_five_nine_years+Schooling+
             I(Hepatitis_B^2)+I(Measles^2)+I(BMI^2)+I(Incidents_HIV^2)+I(GDP_per_capita^2)+I(Population_mln^2)+I(Thinness_ten_nineteen_years^2)+I(Thinness_five_nine_years^2)+I(Schooling^2)
            ,data = data)
#Complete 2nd order
```

Model4: complete second order

Complete 2nd order

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9 +$$
$$\beta_{10} x_1 x_2 + \beta_{11} x_1 x_3 + \beta_{12} x_1 x_4 + \beta_{13} x_1 x_5 + \beta_{14} x_1 x_6 + \beta_{15} x_1 x_7 + \beta_{16} x_1 x_8 + \beta_{17} x_1 x_9 +$$
$$\beta_{18} x_2 x_3 + \beta_{19} x_2 x_4 + \beta_{20} x_2 x_5 + \beta_{21} x_2 x_6 + \beta_{22} x_2 x_7 + \beta_{23} x_2 x_8 + \beta_{24} x_2 x_9 +$$
$$\beta_{25} x_3 x_4 + \beta_{26} x_3 x_5 + \beta_{27} x_3 x_6 + \beta_{28} x_3 x_7 + \beta_{29} x_3 x_8 + \beta_{30} x_3 x_9 +$$
$$\beta_{31} x_4 x_5 + \beta_{32} x_4 x_6 + \beta_{33} x_4 x_7 + \beta_{34} x_4 x_8 + \beta_{35} x_4 x_9 +$$
$$\beta_{36} x_5 x_6 + \beta_{37} x_5 x_7 + \beta_{38} x_5 x_8 + \beta_{39} x_5 x_9 +$$
$$\beta_{40} x_6 x_7 + \beta_{41} x_6 x_8 + \beta_{42} x_6 x_9 +$$
$$\beta_{43} x_7 x_8 + \beta_{44} x_7 x_9 +$$
$$\beta_{45} x_8 x_9 +$$
$$\beta_{46} x_1^2 + \beta_{47} x_2^2 + \beta_{48} x_3^2 + \beta_{49} x_4^2 + \beta_{50} x_5^2 + \beta_{51} x_6^2 + \beta_{52} x_7^2 + \beta_{53} x_8^2 + \beta_{54} x_9^2$$

```
model4 <- lm(Life_expectancy ~ Hepatitis_B + Measles + BMI + Incidents_HIV + GDP_per_capita + Population_mln + Thinness_ten_nineteen_years + Thinness_five_nine_years + Schooling +
               Hepatitis_B * Measles + Hepatitis_B * BMI + Hepatitis_B * Incidents_HIV + Hepatitis_B * GDP_per_capita + Hepatitis_B * Population_mln +
               Hepatitis_B * Thinness_ten_nineteen_years + Hepatitis_B * Thinness_five_nine_years + Hepatitis_B * Schooling +
               Measles * BMI + Measles * Incidents_HIV + Measles * GDP_per_capita + Measles * Population_mln + Measles * Thinness_ten_nineteen_years + Measles * Thinness_five_nine_years +
               Measles * Schooling +
               BMI * Incidents_HIV + BMI * GDP_per_capita + BMI * Population_mln + BMI * Thinness_ten_nineteen_years + BMI * Thinness_five_nine_years + BMI * Schooling +
               Incidents_HIV * GDP_per_capita + Incidents_HIV * Population_mln + Incidents_HIV * Thinness_ten_nineteen_years + Incidents_HIV * Thinness_five_nine_years + Incidents_HIV * Schooling +
               GDP_per_capita * Population_mln + GDP_per_capita * Thinness_ten_nineteen_years + GDP_per_capita * Thinness_five_nine_years + GDP_per_capita * Schooling +
               Population_mln * Thinness_ten_nineteen_years + Population_mln * Thinness_five_nine_years + Population_mln * Schooling +
               Thinness_ten_nineteen_years * Thinness_five_nine_years + Thinness_ten_nineteen_years * Schooling +
               I(Hepatitis_B^2) + I(Measles^2) + I(BMI^2) + I(Incidents_HIV^2) + I(GDP_per_capita^2) + I(Population_mln^2) + I(Thinness_ten_nineteen_years^2)
             + I(Thinness_five_nine_years^2) + I(Schooling^2),
             data = data)
```

ANOVA - Nested F Test

Anova Test	**Hypothesis**	**Result**
Complete 2nd Vs. 2nd order Squared only Anova(model4, model2)	H0: B10=…B44=B45=0 H1: at least one of the interaction term's B's is nonzero	P value = 2.2e-16 *** < alpha = 0.05 Reject H0, at least one of the interaction terms Betas is nonzero
Complete 2nd Vs. 2nd order interaction only Anova(model4, model3)	H0: B46=…B53=B54=0 H1: at least one of the squared terms B's is Nonzero	P value = 2.2e-16 *** < alpha = 0.05 Reject H0, at least one of the squared terms is nonzero
Complete 2nd Vs. Complete First Anova(model4, model1)	H0: B10=…B53=B54=0 H1: at least one of the second order terms is Nonzero	P value = 2.2e-16 *** < alpha = 0.05 Reject H0, at least one of the second order terms is nonzero

Model 4 wins!

anova(model4, model2) - **model4 wins**

```
    I(Population_mln^2) + I(Thinness_ten_nineteen_years^2) +
    I(Thinness_five_nine_years^2) + I(Schooling^2)
  Res.Df   RSS  Df Sum of Sq      F    Pr(>F)
1   2810 27117
2   2845 34952 -35   -7834.3 23.195 < 2.2e-16 ***
---
```

anova(model4, model3) - **model4 wins**

```
    Schooling
  Res.Df   RSS Df Sum of Sq      F    Pr(>F)
1   2810 27117
2   2819 32710 -9     -5593 64.396 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
> anova(model4, model1)
```

anova(model4, model1) - **model4 wins**

```
    I(Schooling^2)
Model 2: Life_expectancy ~ Hepatitis_B + Measles + BMI + Incidents_HIV +
    GDP_per_capita + Population_mln + Thinness_ten_nineteen_years +
    Thinness_five_nine_years + Schooling
  Res.Df   RSS  Df Sum of Sq      F    Pr(>F)
1   2810 27117
2   2854 51619 -44    -24502 57.704 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
>           Incidents_HIV*Schooling+
```

Statistically evaluating model utility - Global F test:

H0 : B1 = B2….= B53 = B54 = 0
Ha : at least one Bi /= 0

Reject H0

The extremely small p value indicates that the model is statistically adequate (<0.05). The additional predictors significantly improve the fit.

```
Hepatitis_B:GDP_per_capita                          -4.694e-07  4.480e-07  -1.048 0.294836
Hepatitis_B:Population_mln                           9.921e-06  3.961e-05   0.250 0.802224
Hepatitis_B:Thinness_ten_nineteen_years              1.100e-03  3.003e-03   0.366 0.714074
Hepatitis_B:Thinness_five_nine_years                -1.210e-03  2.948e-03  -0.410 0.681471
Hepatitis_B:Schooling                               -3.443e-03  2.320e-03  -1.484 0.137987
Measles:BMI                                          1.080e-02  2.693e-03   4.010 6.24e-05 ***
Measles:Incidents_HIV                               -1.288e-02  2.142e-03  -6.014 2.04e-09 ***
Measles:GDP_per_capita                               3.975e-07  5.188e-07   0.766 0.443633
Measles:Population_mln                              -9.298e-05  6.420e-05  -1.448 0.147671
Measles:Thinness_ten_nineteen_years                  1.973e-04  2.733e-03   0.072 0.942437
Measles:Thinness_five_nine_years                     2.371e-03  2.732e-03   0.868 0.385552
Measles:Schooling                                   -1.740e-02  2.413e-03  -7.209 7.21e-13 ***
BMI:Incidents_HIV                                    8.749e-02  2.863e-02   3.056 0.002263 **
BMI:GDP_per_capita                                  -1.862e-05  4.058e-06  -4.588 4.67e-06 ***
BMI:Population_mln                                  -3.428e-03  9.584e-04  -3.577 0.000353 ***
BMI:Thinness_ten_nineteen_years                     -5.667e-02  4.405e-02  -1.287 0.198352
BMI:Thinness_five_nine_years                         2.819e-02  4.327e-02   0.652 0.514740
BMI:Schooling                                       -2.727e-02  2.698e-02  -1.011 0.312129
Incidents_HIV:GDP_per_capita                         4.576e-05  1.565e-05   2.924 0.003485 **
Incidents_HIV:Population_mln                        -1.494e-02  2.710e-03  -5.514 3.83e-08 ***
Incidents_HIV:Thinness_ten_nineteen_years            4.586e-02  2.638e-02   1.738 0.082324 .
Incidents_HIV:Thinness_five_nine_years              -3.855e-02  2.579e-02  -1.495 0.135077
Incidents_HIV:Schooling                              9.258e-02  2.170e-02   4.267 2.05e-05 ***
GDP_per_capita:Population_mln                        3.565e-07  1.353e-07   2.635 0.008459 **
GDP_per_capita:Thinness_ten_nineteen_years          -8.049e-05  2.673e-05  -3.011 0.002627 **
GDP_per_capita:Thinness_five_nine_years              6.527e-05  2.798e-05   2.332 0.019745 *
GDP_per_capita:Schooling                             5.498e-06  3.775e-06   1.456 0.145376
Population_mln:Thinness_ten_nineteen_years           2.649e-03  9.414e-04   2.814 0.004923 **
Population_mln:Thinness_five_nine_years             -3.512e-03  8.955e-04  -3.922 9.01e-05 ***
Population_mln:Schooling                            -1.369e-03  9.758e-04  -1.403 0.160805
Thinness_ten_nineteen_years:Thinness_five_nine_years -7.683e-04  5.809e-03  -0.132 0.894794
Thinness_ten_nineteen_years:Schooling                2.928e-02  1.071e-02   2.733 0.006314 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.106 on 2810 degrees of freedom
Multiple R-squared:  0.8929,    Adjusted R-squared:  0.8909
F-statistic: 442.2 on 53 and 2810 DF,  p-value: < 2.2e-16
```

Statistically evaluating model utility - Global F test:

Residual Standard Error:
The complete second order model can predict life expectancy within 2s = 6.216 years of its true value.

Coefficient of Variation:
CV = 4.51
Relatively low, less than 10%

Adjusted R^2:
89.09% of the variance of life expectancy data is explained by the complete 2nd order model.

```
Residual standard error: 3.106 on 2810 degrees of freedom
Multiple R-squared:  0.8929,    Adjusted R-squared:  0.8909
F-statistic: 442.2 on 53 and 2810 DF,  p-value: < 2.2e-16
```

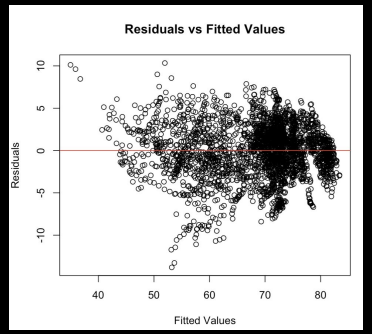
```
> mean(data$Life_expectancy)
[1] 68.85608
```

```
> CV <- 100*(s/68.85608)
> CV
[1] 4.510858
```

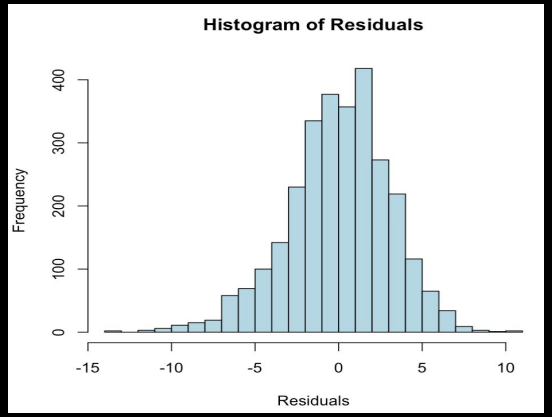
Equation of Complete 2nd Order Model

$$
\begin{aligned}
E(y) = {} & 52.09 + 0.4113x_1 - 0.1623x_2 + 6.379x_3 - 5.2x_4 + 0.0008208x_5 + 0.106x_6 + 0.9998x_7 + 0.9998x_8 \\
& - 1.039x_9 + \\
& 4.133x_1^2 + 0.0004587x_2^2 + 0.0007303x_3^2 - 0.1015x_4^2 + 0.08948x_5^2 - 0.000000002951x_6^2 - 0.000003423x_7^2 \\
& - 0.007381x_8^2 + 0.04249x_9^2 \\
& - 0.09929x_1x_2 - 0.0006561x_1x_3 - 0.01333x_1x_4 + 0.00586x_1x_5 - 0.0000004694x_1x_6 + 0.000009921x_1x_7 + 0.0011x_1x_8 \\
& - 0.00121\beta_{17}x_1x_9 + \\
& - 0.003443x_2x_3 + 0.0108x_2x_4 - 0.01288x_2x_5 + 0.0000003975x_2x_6 - 0.00009298x_2x_7 + 0.0001973x_2x_8 + 0.002371x_2x_9 + \\
& - 0.0174x_3x_4 + 0.08749x_3x_5 - 0.00001862x_3x_6 - 0.003428x_3x_7 - 0.05667x_3x_8 + 0.02819x_3x_9 + \\
& - 0.02727x_4x_5 + 0.00004576x_4x_6 - 0.01494x_4x_7 + 0.04586x_4x_8 - 0.03855x_4x_9 + \\
& 0.09258x_5x_6 + 0.0000003565x_5x_7 - 0.00008049x_5x_8 + 0.00006527x_5x_9 + \\
& 0.000005498x_6x_7 + 0.002649x_6x_8 - 0.003512x_6x_9 + \\
& - 0.001369x_7x_8 - 0.0007683x_7x_9 + \\
& 0.02928x_8x_9
\end{aligned}
$$

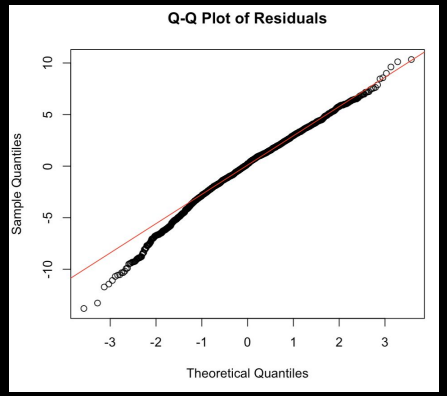
Residual Analysis



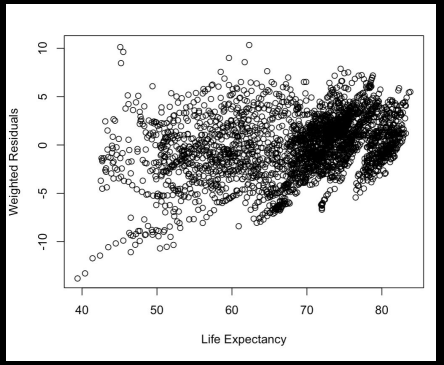
- Random scatter ✅
- Subtle signs of nonlinearity as there is more spread in lower-fitted values than higher-fitted values ❗ but nothing major



- Follows a normal distribution and centering around 0 ✅
- Approximately normally distributed



- Mainly close to straight line – normality assumption holds ✅
- Tails deviate ❗ but not significant enough to invalidate model



- Indicates mild heteroscedasticity because the variance of residuals is not perfectly uniform

Conclusion: Prediction at the Country level

- **Dependent variable**: Life_expectancy
- **Independent variables (9)**:
  Hepatitis_B, Measles, BMI, Incidents_HIV, GDP_per_capita, Population_mln, Thinness_ten_nineteen_years, Thinness_five_nine_years, Schooling